# Symbolic Verification Coverage for the $(1, 1, 3)$ CMF Proof

Executable certificate inventory

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## Purpose and methodology

This document maps the mathematical derivation to `verify_formula_2_1_derivation.py`. The proof starts with the Gauss-hypergeometric conservative matrix field (CMF), follows the trajectory $(1, 1, 3)$, changes to a consecutive-value basis by a rational coboundary, rescales the scalar recurrence to the printed polynomial continued fraction, and then uses the Euler-integral minimal-solution method followed by Pincherle's theorem.

The verifier delegates the CMF and recurrence construction to `ramanujantools`. It uses `pFq` to generate the field, `CMF.trajectory_matrix` to restrict it to the chosen line, `Matrix.companion_coboundary_matrix` to find the change of basis, and `PCF.deflate_all` to obtain the compact challenge coefficients.

**Exact** means that the script checks an identity over exact SymPy expressions or evaluates an exact definite integral. **Partial** means that the specialized algebra or theorem hypotheses are checked but a cited analytic result remains necessary. **Analytic** marks a genuinely analytic argument in the written proof. Numerical checks are independent diagnostics, not substitutes for proof.

| Proof step | Executable certificate | Coverage |
|---|---|---|
| Printed $a_n, b_n$ | `raw_trajectory_pcf` calls `PCF(trajectory_matrix)`, and `challenge_pcf` calls `PCF.deflate_all`. `pcf_coefficients` exposes the resulting polynomials and compares them with `expected_pcf_coefficients`, which stores the printed target. | **Exact** |
| CMF basis matrices | `hypergeometric_cmf` calls `pFq(2,1,1/2)`, and `cmf_basis_matrices` obtains $M_x, M_y, M_z$ through `CMF.M`. The wrapper only renames the internal axes to $x, y, z$. | **Partial** |
| CMF conservation | `cmf_conservation_validation` calls `CMF.validate_conserving`, which checks every coordinate pair. | **Exact** |
| Trajectory point and path | `library_trajectory`, `library_start_position`, and `trajectory_position` call `CMF.trajectory_substitution` to construct $p + n(1, 1, 3)$. | **Exact** |
| Closed trajectory matrix | `trajectory_matrix_113` calls `CMF.trajectory_matrix`. `expected_trajectory_matrix_113` and the `trajectory_matrix` check compare all four rational entries. | **Exact** |
| Meaning of the change of basis | `companion_coboundary_matrix_113` calls `Matrix.companion_coboundary_matrix` to find $U(n)$, while `coboundary_basis_residual` checks that the second column of $U(n)$ is the first column of $T(n)$. This is the coefficient identity behind $[F_n, G_n]U(n) = [F_n, F_{n+1}]$. | **Exact** |
| Coboundary identity | `companion_matrix_113` calls `Matrix.as_companion`. `expected_companion_matrix_113` and `coboundary_identity` call `Matrix.coboundary` and check $$T(n)U(n+1) = U(n)C(n).$$ | **Exact** |
| Scalar recurrence | `coboundary_point_checks` adds checks at five regular indices. `companion_recurrence_coefficients` extracts $A(n), B(n)$ from the library companion matrix. `hypergeometric_recurrence_point_checks` independently expands $F_n$ and checks $$F_{n+2} = A(n)F_{n+1} + B(n)F_n$$ at three regular indices. | **Partial**: the general shift action remains cited |
| Scaling to the PCF | `minimal_solution_scaling_ratio` and the `minimal_solution_a`, `minimal_solution_b` checks verify $r_n A(n) = a_n$ and $r_n r_{n-1} B(n) = b_n$. The function derives $r_n$ from the library-generated $a_n/A(n)$; `initial_scaling_ratio` checks $r_0 = 24/77$. | **Exact** |

| Proof step | Executable certificate | Coverage |
|---|---|---|
| Raw normalization and deflation | `raw_trajectory_pcf` uses the `PCF(Matrix)` constructor; `challenge_pcf` applies `PCF.deflate_all`. `raw_pcf_coefficients` exposes the first object's coefficients, and `deflation_factor` recovers the removed factor from both objects. | **Exact** |
| Euler-integral specialization | `euler_integral_components` and `euler_integrand_factorization` verify that the specialized integrand is $h(t)^n\psi(t)$. Euler's integral representation itself is a cited theorem. | **Partial** |
| Saddle and unique maximum | `saddle_point_data` and `saddle_point_certificate` check the factorization of $h'(t)$, the location of both critical roots, both endpoint values, $h(t_*) > 0$, and the exact value $h(t_*) = -11 + 5\sqrt{5}$. | **Partial** |
| Endpoint estimate | `endpoint_bound_closed_form` verifies the closed form $$\frac{\sqrt{2}}{2n}\left(\frac{1}{8}\right)^n,$$ and `endpoint_base_below_saddle` checks $1/8 < h(t_*)$. The pointwise inequalities producing the bound remain in the proof. | **Partial** |
| Gamma-prefactor growth | `gamma_prefactor_ratio` computes the exact consecutive quotient $$\frac{P_{n+1}}{P_n} = \frac{3(6n+1)(6n+5)}{8n(2n+1)} \longrightarrow \frac{27}{4}.$$ The sharper square-root factor and error term use Stirling's formula. | **Partial** |
| Limiting recurrence and roots | `limiting_companion_matrix`, `companion_characteristic_roots`, and the characteristic-root checks verify the limiting matrix, both roots, their signs, and their distinct moduli. | **Exact** |
| Identification of the small root | `saddle_point_data` and `small_root` verify $$\frac{27}{4}\,(-11 + 5\sqrt{5}) = \frac{-297 + 135\sqrt{5}}{4}.$$ | **Exact** |
| Poincaré–Perron and minimality | The script verifies finite coefficient limits, nonzero limiting $B$, distinct root moduli, positivity-compatible root signs, and the Euler exponential base. The theorem and the growth argument are not formalized. | **Analytic** |
| Transport of minimality | `transport_quotient_residual` verifies cancellation of the common scale in $X_n/Y_n = F_{n+2}/D_{n+2}$; the coefficient transport is checked separately. | **Partial** |
| Pincherle hypotheses and sign | `pcf_numerator_zeros` solves the library-generated $b_n$ and shows that its zeros are $-6/5, -1/2, 0, 4/5$, hence none is a positive integer. `riccati_identity_residual` checks $$R_{n-1} = \frac{b_n}{a_n + R_n}, \qquad R_n = -\frac{Z_n}{Z_{n-1}}.$$ Pincherle's convergence theorem remains cited. | **Partial** |
| Anchor substitutions | `anchor_substitution_certificates` checks the complete Jacobian and both rational integrands after $t = 2u^2/(1+u^2)$. | **Exact** |
| Anchor basis integrals | `anchor_basis_integrals` generates $I_1, \ldots, I_4$ from the integration-by-parts recurrence; `anchor_basis_integrals` in `symbolic_verification` also compares each value with its exact definite integral. | **Exact** |
| Anchor partial fractions and values | `anchor_integral_certificates`, the decomposition checks, the exact integral checks, and `anchor_f1`, `anchor_f2` compare both closed forms with values computed by `pFq.evaluate`. | **Exact** |
| Tail and final simplification | `symbolic_tail` and `pincherle_tail` check $$-\frac{X_0}{X_{-1}} = \frac{6(22 - 7\pi)}{3 - \pi}.$$ `symbolic_limit` and `closed_form_limit` then check the full value $6/(3-\pi)$. | **Partial** |
| Convergence-rate invariant | `companion_characteristic_roots`, `pcf_characteristic_roots`, and `root_ratio_invariant` verify that recurrence normalization preserves the exact root-modulus ratio. The conversion of that ratio into a convergent-error asymptotic is analytic. | **Partial** |
| Independent numerical convergence | `numerical_verification` calls `PCF.limit` on `challenge_pcf` and compares the exact rational convergent with $6/(3-\pi)$ at arbitrary precision. | Numerical diagnostic |